

GROUP 1 — DATE FEATURES (6 FEATURES) — FROM TODAY'S DATE, NO HISTORY NEEDED

1

day_of_week

1 = Tuesday

df['date'].dt.dayofweek

0=Mon 1=Tue 2=Wed 3=Thu 4=Fri 5=Sat 6=Sun

Tells model what day of week it is. Model learned from Kaggle that weekends have higher sales than weekdays for most products.

Dec 31 2024 = Tuesday → day_of_week = 1

2

month

12 = December

df['date'].dt.month

1=Jan 6=Jun 12=Dec

Most important seasonal signal. Model learned December = high demand, January = low demand from 913k Kaggle rows.

Dec 31 2024 → month = 12 → model predicts HIGH

3

quarter

4 = Q4

df['date'].dt.quarter

Q1=Jan-Mar Q2=Apr-Jun Q3=Jul-Sep Q4=Oct-Dec

Q4 is festive season globally. Model learned Q4 consistently shows higher demand than Q1 across all product types.

December → quarter = 4 → festive boost applied

4

is_weekend

0 or 1

(day_of_week >= 5).astype(int)

Sat or Sun = 1, else = 0

Binary flag. Weekend shopping patterns differ from weekdays. Simple but powerful signal for retail demand.

Tuesday → is_weekend = 0 (no weekend boost)

5

is_month_start

0 or 1

(date.day <= 5).astype(int)

Day 1-5 = 1, else = 0

First 5 days of month = salary week. People buy more after receiving salary. Especially strong for FMCG and electronics.

Dec 31 → day=31 → is_month_start = 0

6

is_month_end

0 or 1

(date.day >= 25).astype(int)

Day 25-31 = 1, else = 0

Last week of month = bulk buying before month closes. Businesses reorder supplies. Strong signal for B2B products.

Dec 31 → day=31 → is_month_end = 1 (boost!)

GROUP 2 — LAG FEATURES (4 FEATURES) — PAST SALES VALUES, NEEDS HISTORY

7

lag_7

sales 7 days ago = 48

df['sales'].shift(7)

If today is Dec 31 → lag_7 = sales on Dec 24

The single most important feature. Recent past strongly predicts near future. If last week was busy, next week likely busy too.

Visual — how lag_7 is picked from timeline:

Dec 17

Dec 18

Dec 19

Dec 20

Dec 21

Dec 22

Dec 24 ← lag_7

Dec 25

Dec 26

Dec 27

Dec 31 TODAY

lag_7 = value at Dec 24 = 48 units. Red bar = the lag_7 value picked.

8

lag_14

44

df['sales'].shift(14)

sales 14 days ago

2 weeks ago sales. Helps model understand 2-week buying cycles common in B2B warehouses.

Dec 31 → Dec 17 sales = 44

9

lag_30

41

df['sales'].shift(30)

sales 30 days ago

Last month's sales. Used with lag_7 to compute trend — is demand rising or falling over the month?

Dec 31 → Dec 1 sales = 41

10

lag_365

31

df['sales'].shift(365)

same day last year

Most powerful seasonal feature. Compares today to exact same day 1 year ago. Captures product-specific seasonal patterns from user's own data.

Dec 31 2024 → Dec 31 2023 sales = 31.
Needs 2 years CSV.

GROUP 3 — ROLLING FEATURES (3 FEATURES) — AVERAGES OVER A WINDOW

11

rolling_mean_7

44.2

12

rolling_mean_30

39.7

13

rolling_std_7

4.1

```
sales.shift(1).rolling(7).mean()  
avg of last 7 days
```

Smooths out daily noise. If one day was unusually high or low, the average gives the true recent trend. Very stable signal.

[40,42,44,45,47,48,44] → avg = 44.2

```
sales.shift(1).rolling(30).mean()  
avg of last 30 days
```

Monthly baseline. Compared with rolling_mean_7 — if 7-day avg is higher than 30-day avg, demand is accelerating upward.

rm7=44.2 > rm30=39.7 → demand accelerating

```
sales.shift(1).rolling(7).std()  
standard deviation last 7 days
```

Measures volatility. High std = unpredictable sales. Low std = stable. Model uses this to adjust confidence of prediction.

std=1.2 → stable. std=12.4 → volatile, lower confidence.

GROUP 4 — DERIVED FEATURES (2 FEATURES) — COMPUTED FROM OTHER FEATURES

14 trend_direction

+7.0 rising

```
lag_7 - lag_30  
48 - 41 = +7
```

Positive = demand rising. Negative = demand falling. Simple but extremely powerful. Model gives high weight to this feature.

Rising (+7)
lag7=48 > lag30=41
predict HIGH demand

Falling (-8)
lag7=32 < lag30=40
predict LOW demand

15 yoy_growth

+0.61 = +61%

```
(lag_7 - lag_365) / (lag_365 + 1)  
(48 - 31) / (31 + 1) = 0.53
```

Year-over-year growth rate. Positive means this year is stronger than last year same period. The +1 in denominator prevents division by zero.

+0.61 growth
61% stronger than last year
→ model predicts higher

-0.05 decline
5% weaker than last year
→ model predicts lower

HOW ALL 15 FEATURES WORK TOGETHER — REAL EXAMPLE

Date signals

month = 12 → December
quarter = 4 → festive
is_month_end = 1 → bulk buying
is_weekend = 0 → weekday
Combined → moderate seasonal boost

History signals

lag_7 = 48 → recent high
lag_30 = 41 → month was lower
lag_365 = 31 → last year was low
rolling_mean_7 = 44.2 → rising
Combined → strong upward momentum

Trend signals

trend_direction = +7 → rising
yoy_growth = +0.61 → 61% up
rolling_std = 4.1 → stable
rm7 > rm30 → accelerating
Combined → high confidence spike

Model sees all 15 values together → multiplies each by its learned weight → passes through 3 neural network layers → outputs 0.87 → scaled back to 343 units next week